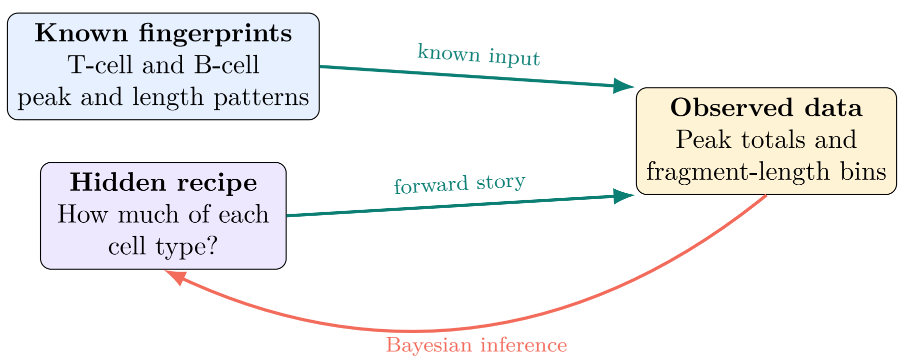
Bayesian Models from Scratch

A high-school companion to ShapeMix-ATAC

No calculus required: pictures, small-number examples, and equation-reading practice



Big idea

This tutorial prepares you to read `ShapeMix_ATAC_Bayesian_Model_Tutorial.pdf`. You will learn what a model is, how Bayes' rule updates a belief, why counts need probability distributions, how mixtures are represented, and what a MAP estimate does. Every new symbol is translated into ordinary language.

Companion tutorial, version 1.0

August 2026

Contents

1 How to use this tutorial

This document is a *prequel*, not a replacement for the full ShapeMix tutorial. Read it in order the first time. Keep a pencil nearby and actually try the short checkpoints before reading their answers.

If you want to learn ...	Focus on ...
Bayes' rule itself	Sections 3–4
Why special distributions are used	Section 5
How mixtures are inferred	Sections 6 and 9
The ShapeMix symbols and equations	Sections 8–10
What MAP means	Section 12
How to approach the original PDF	Section 14

1.1 What you need before starting

You should be comfortable with fractions, percentages, multiplication, and simple algebra. Summation notation such as $\sum$ will be explained. You do not need calculus, programming, or previous statistics.

Check your understanding

By the end, you should be able to explain this sentence:

> ShapeMix combines a negative-binomial model for the number of cut sites in a peak, a conditional multinomial model for how those cuts split among parent-fragment length bins, and a Gamma prior on positive cell-type abundances; it reports the MAP mixture.

2 Start with the scientific question

2.1 What is observed, and what is hidden?

In spatial ATAC-seq, a tissue location can contain material from several cell types. The experiment gives us counts at that location, but it does not label each count with the cell type that produced it.

Role	ShapeMix example
Observed data	Cut-site counts in genomic peaks, grouped by the length of each cut site's parent fragment.
Known inputs	Reference fingerprints learned from labeled training cells.
Hidden quantity	The amount of each cell type in the mixed spot.
Reported answer	The estimated cell-type proportions.

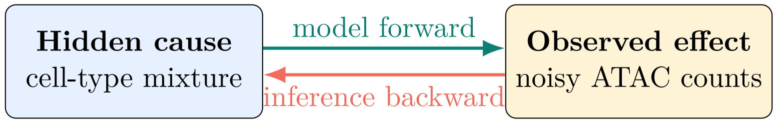


Figure 1: A model describes how a hidden cause could produce an observed effect. Inference asks which hidden causes are supported by the effect we actually observed.

2.2 A model is a precise story

A **model** is not the data and it is not the answer. It is a set of rules that connect possible hidden explanations to possible observations.

For ShapeMix, the story is approximately:

1. A spot has some effective amount of each cell type.
2. Each cell type has a known reference fingerprint across peaks and length bins.
3. Mixing those fingerprints gives expected peak totals and expected length proportions.
4. Random experimental variation makes the observed counts differ from those expectations.

Big idea

A Bayesian model first tells a story in the easy direction:

$$\text{hidden mixture} \longrightarrow \text{expected signal} \longrightarrow \text{noisy data.}$$

Bayesian inference uses the observed data to reason in the reverse direction.

2.3 One ATAC counting detail

A deduplicated ATAC fragment has two ends, corresponding to two Tn5 cut-site observations. Each end is assigned independently to a peak. Therefore one fragment can contribute two cuts to one peak, one cut to each of two peaks, one selected cut, or no selected cuts.

For a simple classroom picture, we can assume both ends of every drawn fragment land in the same peak. Under that assumption:

$$12 \text{ fragments} \times 2 \text{ ends per fragment} = 24 \text{ cut-site counts.}$$

Watch out

ShapeMix's primary count unit is a *cut site tagged by the length of its parent fragment*, not a unique fragment. Keeping that distinction straight prevents a factor-of-two counting mistake.

3 Probability is a language for uncertainty

3.1 Probabilities are numbers from 0 to 1

A probability of 0 means impossible, 1 means certain, and 0.60 means 60%. Imagine 100 equally likely cards: 60 teal cards and 40 gray cards.

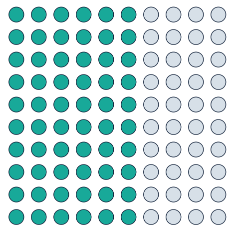


Figure 2: Probability can be pictured as a fraction of equally likely possibilities.

The notation $P(A)$ means the probability of event A. For example,

$$P(\text{short parent fragment}) = 0.60.$$

3.2 Conditional probability changes the question

The vertical bar in $P(A \mid B)$ is read as "given." It means:

> What is the probability of A, if we already know B?

Suppose a reference tells us:

Parent-fragment category	T-cell cut	B-cell cut
Short	80%	20%
Long	20%	80%

Then

$$P(\text{short} \mid T) = 0.80, \qquad P(\text{short} \mid B) = 0.20.$$

These numbers answer: "If the cut came from this cell type, how likely is a short parent fragment?"

Watch out

$$P(\text{short} \mid T) \neq P(T \mid \text{short}).$$

The first predicts data from a known cell type. The second infers a cell type from observed data. Bayes' rule connects the two directions.

3.3 A probability tree

Assume a cut is equally likely to come from a T cell or a B cell before we inspect its parent-fragment length:

$$P(T) = 0.50, \qquad P(B) = 0.50.$$

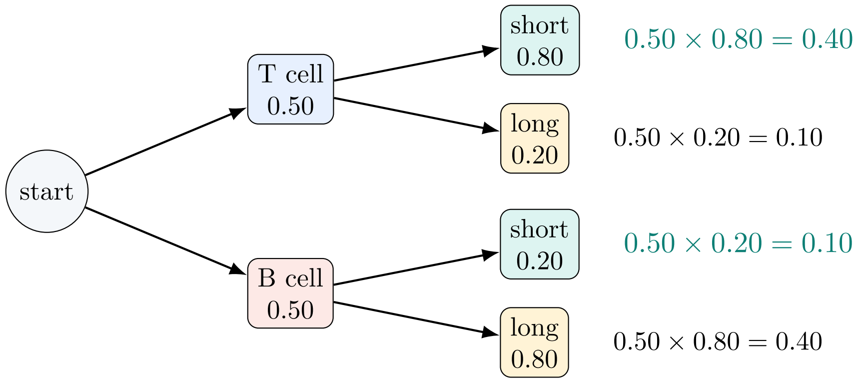


Figure 3: Multiply along a branch. The two highlighted branches are the ways to observe a short-parent cut.

If we observe a short-parent cut, only the two short branches remain. Their weights are 0.40 for T and 0.10 for B. Normalize them so they add to one:

$$P(T \mid \text{short}) = \frac{0.40}{0.40 + 0.10} = 0.80.$$

Check your understanding

Why is the answer 80% rather than 40%? Because 40% is the probability of the *joint event* “T cell and short.” Once short has already been observed, we renormalize within the short branches only.

4 Bayes’ rule: prior, likelihood, and posterior

4.1 The formula and its translation

Bayes’ rule

$$\boxed{P(H \mid E) = \frac{P(E \mid H)P(H)}{P(E)}}$$

$$\underbrace{P(H \mid E)}_{\text{posterior}} = \frac{\underbrace{P(E \mid H)}_{\text{likelihood}}\,\underbrace{P(H)}_{\text{prior}}}{\underbrace{P(E)}_{\text{evidence / normalizer}}}.$$

Here H means a possible hidden explanation, or **hypothesis**, and E means the observed **evidence**.

Word	Question	Short-fragment example
Prior	What was plausible before these data?	The next cut is equally likely to come from T or B.
Likelihood	If this explanation were true, how well would it predict the data?	T predicts short with 80%; B predicts short with 20%.
Posterior	What is plausible after seeing the data?	Given short, support becomes 80% T and 20% B.
Evidence	How common is the observation across all explanations?	0.40+0.10 = 0.50; it normalizes the answer.

Big idea

Bayes' rule is an update:

$$\boxed{\text{posterior} \propto \text{likelihood} \times \text{prior.}}$$

The symbol $\propto$ means "proportional to." It says the numerator gives relative scores; dividing by the evidence makes the scores add to one.

4.2 A complete numerical update

One observed short-parent cut

Let H_T mean the cut came from a T cell and H_B mean it came from a B cell.

Hypothesis	prior	likelihood of short	unnormalized score
H_T	0.50	0.80	0.40
H_B	0.50	0.20	0.10

The scores total 0.50. Divide each score by 0.50:

$$P(H_T \mid \text{short}) = 0.80, \qquad P(H_B \mid \text{short}) = 0.20.$$

4.3 What if the prior is not equal?

Suppose that, before seeing this cut, we assign a 25% prior probability to a T-cell source:

$$P(H_T) = 0.25, \qquad P(H_B) = 0.75.$$

After one short-parent cut,

$$P(H_T \mid \text{short}) = \frac{0.80(0.25)}{0.80(0.25) + 0.20(0.75)} = \frac{0.20}{0.35} \approx 0.57.$$

The short cut moves support toward T, but the stronger B prior still matters. This is a prior about the *source of a cut*; it equals a source-cell fraction only when the cell types have equal expected cut yields.

Check your understanding

Does a prior force the answer? No. A prior is one part of the score. Repeated strong evidence can overwhelm a weak or moderate prior, while weak data leave more room for the prior to matter.

4.4 Probability versus likelihood

The expression $P(E \mid H)$ has two related readings:

- If H is fixed and possible data E vary, it is a probability model for data.
- If the observed E is fixed and candidate explanations H vary, the same expression is called the **likelihood**; it scores the candidates.

Watch out

A likelihood is not automatically a probability distribution over hypotheses. Multiplying by a prior and normalizing is what produces a posterior distribution over hypotheses.

4.5 Classification is not yet deconvolution

The example above asked which *single* cell type produced one cut. ShapeMix asks a harder question: what *mixture* of cell types produced many cuts? The same logic survives, but the hypothesis becomes a continuous recipe such as 67% T and 33% B.

5 Probability distributions: rules for random outcomes

5.1 Random does not mean patternless

A **random variable** is a number whose value can vary when an experiment is repeated. A **probability distribution** lists or describes how plausible each possible value is.

Suppose a model expects 10 cut sites at a peak. Repeating the experiment does not force the count to be exactly 10 every time. We might observe 8, 11, 14, or another nearby value. The distribution describes this variation.

Distribution	What it describes	ShapeMix role
Poisson	A basic nonnegative count whose variance equals its mean.	Useful comparison and software check.
NegBinomial	A nonnegative count with extra variation beyond Poisson.	Models the total cut sites in a peak.
Multinomial	How a fixed total is divided among categories.	Splits a peak total among short, middle, and long bins.
Gamma	A positive continuous quantity.	Prior for positive effective cell-type abundance.

5.2 The symbol $\sim$

$$X \sim \mathrm{NegBinomial}(\text{mean} = 10, \text{inverse-dispersion} = 2)$$

is read:

> The random count X follows a negative-binomial distribution with these settings.

The symbol $\sim$ does *not* mean approximately equal here. It means "is modeled as a random draw from."

5.3 Poisson versus negative binomial

A Poisson distribution is a simple count model:

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \mu.$$

Real sequencing counts often vary more. The target ShapeMix model uses a negative binomial parameterized by a mean μ and an inverse-dispersion ϕ:

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \mu + \frac{\mu^2}{\phi}.$$

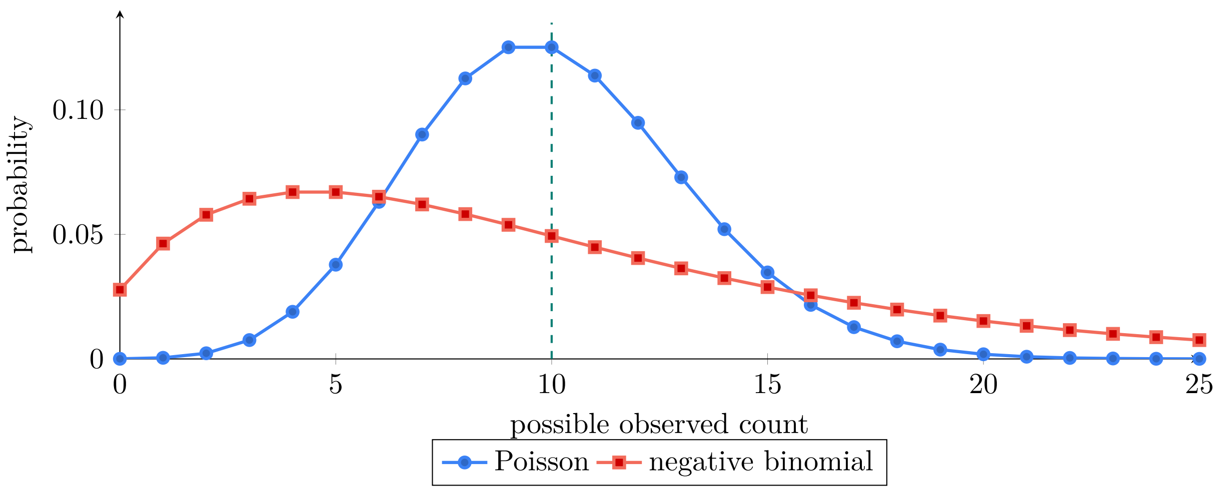


Figure 4: Both examples have mean 10. The negative binomial with $\phi = 2$ is more spread out, so counts farther from 10 remain plausible.

Watch out

Although the target PDF sometimes calls ϕ "dispersion," its formula makes ϕ an *inverse-dispersion*. A larger ϕ makes μ^2/ϕ smaller, so there is *less* extra spread.

Variance calculation

If $\mu = 20$ and $\phi = 5$, then

$$\mathrm{Var}(X) = 20 + \frac{20^2}{5} = 20 + 80 = 100.$$

5.4 Multinomial: divide a fixed total

The multinomial distribution starts after the total N is fixed. Imagine 24 cut-site tokens sorted into three labeled boxes:

The project labels these bins *short*, *mono*, and *long*; in the toy figures, *middle* is a friendlier shorthand for the mono-nucleosomal bin.

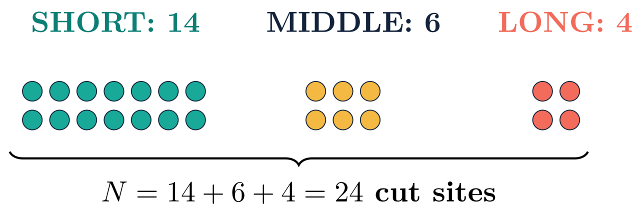


Figure 5: The multinomial asks how one fixed total is allocated across categories. These are the same 24 observations, not 24 new observations.

If the expected bin probabilities are

$$\boldsymbol{\rho} = (0.50,\ 0.30,\ 0.20),$$

then the expected counts for $N = 30$ are

$$30\boldsymbol{\rho} = (15,\ 9,\ 6).$$

The observed split can differ because it is random.

Check your understanding

Which model answers each question?

1. Why was the total count 24 instead of 20 or 30?
2. Given that the total was 24, why did it split 14/6/4?

Answer: the negative binomial addresses question 1; the conditional multinomial addresses question 2.

5.5 Gamma: a distribution on positive values

An effective abundance cannot be negative. A Gamma distribution lives only on positive numbers, so it is a natural prior for abundance.

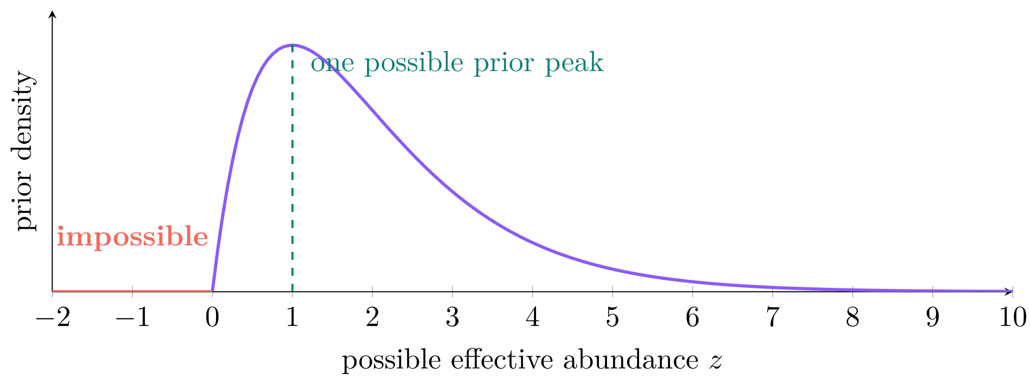


Figure 6: A Gamma density is zero at negative values. The exact curve depends on its hyperparameters; this one is only an illustration.

Big idea

The three distributions have three different jobs:

negative binomial	:	How many cuts?
multinomial	:	How are those cuts divided?
Gamma	:	Which positive abundances were plausible beforehand?

6 A mixture is a weighted recipe

6.1 Reference fingerprints

Start with two simplified reference types at one peak:

Reference	Short	Long	Total
T cell	80%	20%	100
B cell	20%	80%	100

Both types have the same total accessibility in this teaching example. A count-only model sees no difference. Their length patterns, however, are very different.

6.2 Solve a tiny mixture

Suppose an observed spot is 60% short and 40% long. Let π_T be the T-cell proportion. Then the B-cell proportion is $1 - \pi_T$. The expected short fraction is a weighted average:

$$0.60 = 0.80\pi_T + 0.20(1 - \pi_T).$$

Solve:

$$\begin{aligned} 0.60 &= 0.80\pi_T + 0.20 - 0.20\pi_T, \\ 0.40 &= 0.60\pi_T, \\ \pi_T &= \frac{2}{3} \approx 0.67. \end{aligned}$$

So the spot looks like roughly 67% T and 33% B.

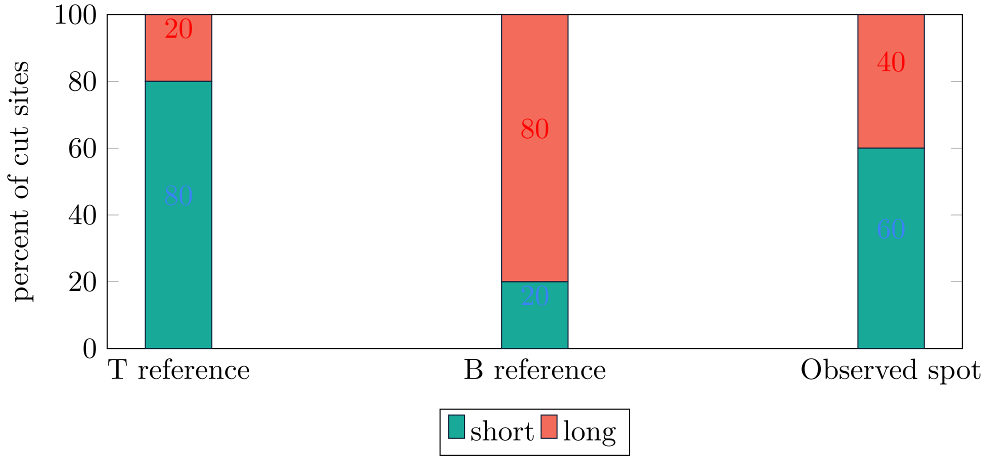


Figure 7: The observed length pattern lies between the two reference patterns. A weighted recipe explains where.

Watch out

This exact algebra treats observed fractions like expected fractions so the idea is easy to see. A probability model is still needed because real observed counts wiggle around their expectations.

Check your understanding

If a spot is 75% T and 25% B, what short fraction is expected?

$$0.75(0.80) + 0.25(0.20) = 0.65.$$

The answer is 65%.

7 The generative story: run the model forward

Generative means the model describes how data could be generated. It does not claim that scientists literally watch nature execute these steps. The steps organize assumptions so they can be checked.

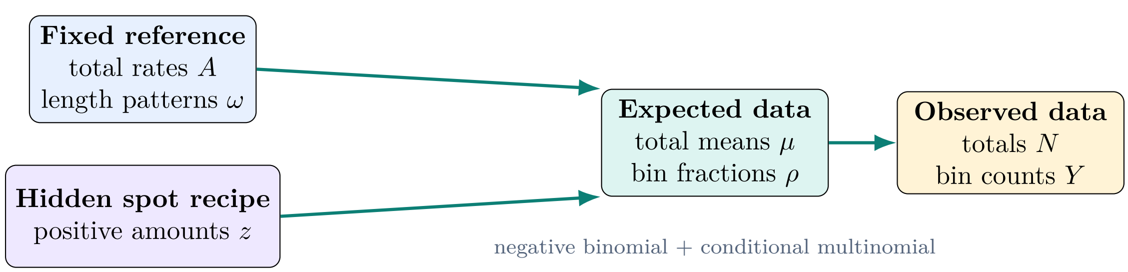


Figure 8: The forward direction. Inference observes the right side and searches for a supported value of z.

One pass through the story is:

1. Choose possible positive cell-type amounts z according to a prior.

2. Combine z with the fixed reference total rates A to calculate an expected peak total μ.
3. Combine z, A, and reference length patterns ω to calculate expected bin fractions ρ.
4. Draw a noisy total N from a negative binomial.
5. Given N, draw the bin counts Y from a multinomial.

Translate the mathematics

Forward question: "If this proposed mixture were true, what data could it produce?"
Inference question: "Among all proposed mixtures, which ones make the observed data plausible and also agree reasonably with the prior?"

7.1 Candidate explanations receive scores

The computer tries candidate abundance values, calculates their expected data, and scores them:

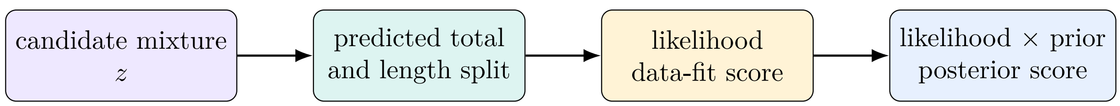


Figure 9: Bayesian computation compares explanations by combining data fit with prior plausibility.

8 Learn to read the notation

Mathematical notation is compressed language. Decode one piece at a time instead of trying to understand an entire equation in one glance.

8.1 Indices are addresses

Index	Address component
s	spatial spot or pseudo-spot
p	genomic peak
b	parent-fragment length bin: short, middle, or long
c	reference cell type

Thus $Y_{3,12,2}$ means:

> the observed cut-site count in spot 3, peak 12, length bin 2.

8.2 Sums, vectors, and dots

$$N_{s,p} = \sum_b Y_{s,p,b}$$

means: hold spot s and peak p fixed, then add the counts over all bins b.

If

$$\boldsymbol{Y}_{s,p,\cdot} = (14, 6, 4),$$

the bold symbol represents the whole three-number vector. The dot means "all values along this index." Its total is $N_{s,p} = 24$.

8.3 A small notation dictionary

Symbol	Read it as	Meaning
$\sum_c$	sum over cell types	Add one contribution from every cell type.
$A \mid B$	A given B	Treat B as already known or fixed.
$\propto$	proportional to	Equal up to a shared normalizing constant.
$\sim$	follows a distribution	A random draw is modeled by the named distribution.
$\mathbb{E}[X]$	expected value of X	The long-run average or model mean.
$\mathrm{Var}(X)$	variance of X	A measure of spread around the mean.
$\log x$	log of x	Turns products into sums and helps numerical stability.
$\arg\max_z f(z)$	z that maximizes f	The input value at the highest point of the function.
$\widehat{z}$	z-hat	An estimated value of z.

8.4 Data, parameters, inputs, and calculated quantities

Table 1: The primary ShapeMix quantities in beginner language.

Symbol	Meaning	Role
$Y_{s,p,b}$	cut sites observed in one length bin	observed data
$N_{s,p}$	all cut sites observed in the peak, $\sum_b Y_{s,p,b}$	observed, calculated from Y
$A_{c,p}$	typical total cut-site rate for reference type c at peak p	fixed reference input
$\omega_{c,p,b}$	typical fraction in bin b for that type and peak	fixed reference input
ϕ_{ref}	reference inverse-dispersion controlling extra total-count noise	fixed reference input
$z_{s,c}$	positive, effective amount of cell type c in spot s	unknown parameter inferred by MAP
e_s	total effective abundance, $\sum_c z_{s,c}$	calculated
$\mu_{s,p}$	expected total cut-site count	calculated
$\rho_{s,p,b}$	expected fraction of cuts in bin b	calculated
$\pi_{s,c}$	normalized cell-type proportion, $z_{s,c}/e_s$	reported output

Watch out

$z_{s,c}$ is not a calibrated nucleus count: it need not be an integer or sum to the number of cells used to simulate a pseudo-spot. In this model it is an effective, depth-scaled reference-cell-equivalent abundance. The primary biological output is the normalized proportion $\pi_{s,c}$.

8.5 Parameter versus hyperparameter

A **parameter** is an unknown value the model infers, such as $z_{s,c}$. A **hyperparameter** is a fixed setting that controls a prior or another piece of model behavior, such as r_z, m_c, or a smoothing strength.

Check your understanding

Is $\mu_{s,p}$ an independently inferred parameter? No. Once z and the fixed reference A are known, μ is calculated deterministically.

9 A complete ShapeMix example with small numbers

9.1 Set up the reference

Use one peak, two cell types, and three length bins. Let both reference types have the same total rate:

$$A_T = A_B = 6 \text{ cuts per effective reference unit.}$$

Their conditional length fingerprints differ:

$$\boldsymbol{\omega}_T = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right), \qquad \boldsymbol{\omega}_B = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right).$$

In words, T-cell cuts tend to have short parents, B-cell cuts tend to have long parents, and both have one-third middle-parent cuts.

9.2 Try one hidden mixture

Suppose the candidate abundance is

$$z_T = 3, \qquad z_B = 1.$$

Its normalized proportions are

$$\pi_T = \frac{3}{3+1} = 0.75, \qquad \pi_B = \frac{1}{3+1} = 0.25.$$

First compute the expected total:

$$\mu = z_T A_T + z_B A_B = 3(6) + 1(6) = 24.$$

Now compute each expected bin count v_b:

$$v_{\text{short}} = 3(6)\left(\frac{1}{2}\right) + 1(6)\left(\frac{1}{6}\right) = 9 + 1 = 10,$$

$$v_{\text{middle}} = 3(6)\left(\frac{1}{3}\right) + 1(6)\left(\frac{1}{3}\right) = 6 + 2 = 8,$$

$$v_{\text{long}} = 3(6)\left(\frac{1}{6}\right) + 1(6)\left(\frac{1}{2}\right) = 3 + 3 = 6.$$

Check conservation:

$$10 + 8 + 6 = 24 = \mu.$$

Finally divide by the total to obtain multinomial probabilities:

$$\boldsymbol{\rho} = \frac{(10, 8, 6)}{24} = \left(\frac{5}{12}, \frac{1}{3}, \frac{1}{4}\right).$$

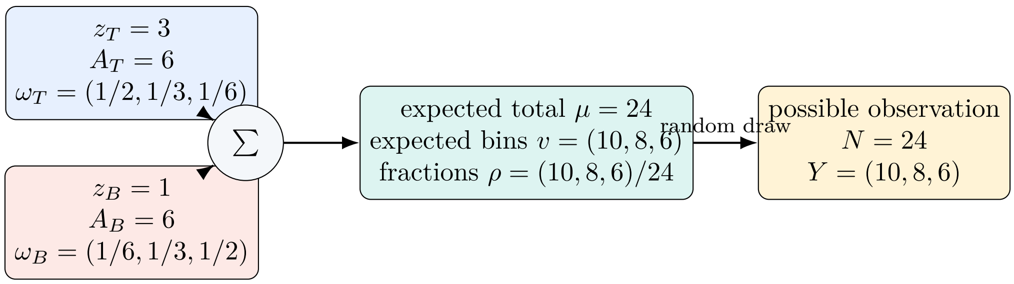


Figure 10: A complete one-peak calculation. The observation happens to equal the expectation here so the arithmetic is easy to follow; real observations fluctuate.

Big idea

Both models observe $N = 24$. With $A_T = A_B$ and total effective abundance $e = z_T + z_B = 4$ held fixed, count-only predicts $\mu = 24$ for every T/B split. ShapeMix also predicts the split $(10, 8, 6)$, which depends on the mixture because $\omega_T \neq \omega_B$.

9.3 Connect back to fragments

If all fragment ends are assigned to this same peak, the observed

$$Y = (10, 8, 6)$$

can be pictured as 5 short, 4 middle, and 3 long fragments. That is 12 fragments and $2 \times 12 = 24$ cut-site counts.

10 The two-part ShapeMix likelihood

10.1 One joint story, factored into two questions

ShapeMix writes the data model as

$$\boxed{p(N, Y \mid z, A, \omega, \phi_{\text{ref}}) = p(N \mid z, A, \phi_{\text{ref}})\, p(Y \mid N, z, A, \omega).}$$

This is the probability product rule: model the total, then model the split conditional on that total.

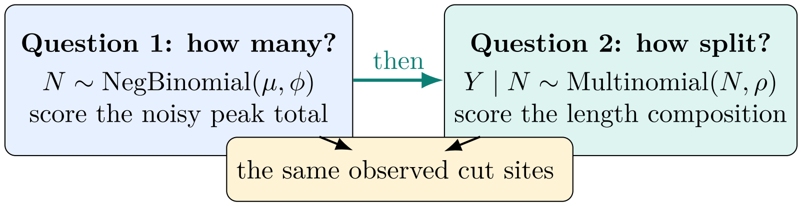


Figure 11: The likelihood separates total-count information from conditional shape information without inventing extra observations.

Watch out

Using N and $Y \mid N$ does not double-count the data. It is a factorization of one joint probability. The negative binomial asks why the sum had its observed value; the conditional multinomial asks how that fixed sum was allocated.

10.2 The two model arms

The count-only objective uses

$$\log p(N \mid z, A, \phi_{\text{ref}}) + \log p(z).$$

ShapeMix uses

$$\log p(N \mid z, A, \phi_{\text{ref}}) + \log p(Y \mid N, z, A, \omega) + \log p(z).$$

Everything except the conditional shape term is kept the same.

Ingredient	Count-only	ShapeMix	Purpose
Negative-binomial total	yes	yes	same peak-count evidence
Conditional multinomial	no	yes	extra length-bin evidence
Gamma prior on z	yes	yes	same regularization
Data, references, optimizer	same	same	fair comparison

Check your understanding

If every cell type has exactly the same ω, can shape help identify the mixture? No. Then ρ does not change with z, so the conditional shape term contains no mixture information.

11 The prior: make assumptions visible

11.1 Why use a prior?

Observed data may be noisy or ambiguous. A prior gives the model a transparent starting preference before the current spot is examined. For ShapeMix, it also enforces an essential fact: abundance must be positive.

The primary model uses

$$\boxed{z_{s,c} \sim \text{Gamma}\left(\text{shape} = r_z,\ \text{rate} = \frac{r_z}{m_c}\right).}$$

Under this **shape–rate** convention,

$$\mathbb{E}[z_{s,c}] = m_c, \qquad \mathrm{Var}(z_{s,c}) = \frac{m_c^2}{r_z}.$$

The general symbols make the prior's roles visible. Under the frozen MVP v1 evaluation contract, $r_z = 2$ and $m_c = 2$ for every cell type, so

$$z_{s,c} \sim \mathrm{Gamma}(\mathrm{shape} = 2, \mathrm{rate} = 1).$$

Translate the mathematics

m_c sets the prior mean for the effective abundance of cell type c. Larger r_z makes the prior more concentrated around that mean; smaller r_z makes it broader. Both are fixed before final evaluation.

Watch out

Some books define a Gamma distribution using a *scale* instead of a *rate*. The target ShapeMix tutorial uses shape and rate. The second argument r_z/m_c is a rate.

11.2 The prior is neither cheating nor certainty

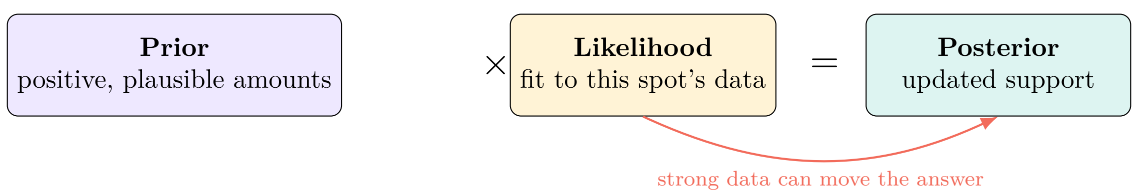


Figure 12: The posterior balances prior plausibility with data fit. Neither component should be hidden.

With little data, many mixtures can fit and the prior has more influence. With strong, consistent data, the likelihood becomes sharply different across candidates and can pull the posterior away from the prior.

Check your understanding

Which is a parameter: $z_{s,c}$ or r_z? $z_{s,c}$ is inferred, so it is a parameter. r_z controls its prior and is fixed, so it is a hyperparameter.

12 Posterior and MAP: find the highest-supported mixture

12.1 A teaching grid of candidate mixtures

Return to the three-bin example:

$$\omega_T = (1/2, 1/3, 1/6), \qquad \omega_B = (1/6, 1/3, 1/2), \qquad Y = (10, 8, 6).$$

To make the update visible, temporarily consider only five possible T-cell proportions and give them equal prior probability. The conditional multinomial likelihood supplies the data-fit score.

Candidate π_T	Expected short/middle/long at $N = 24$	Posterior support on grid
0%	4/8/12	0.3%
25%	6/8/10	5.7%
50%	8/8/8	26.4%
75%	10/8/6	**43.8%**
100%	12/8/4	23.8%

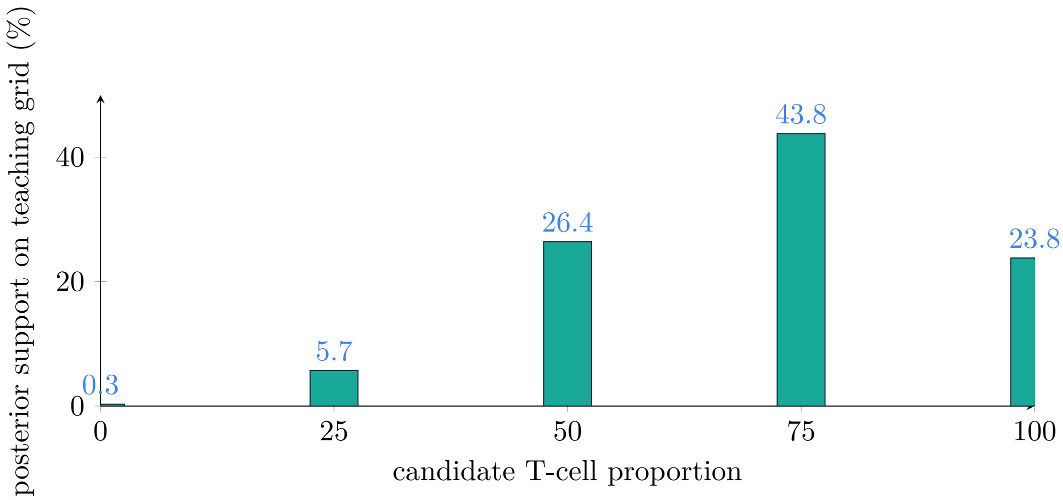


Figure 13: The 75% T candidate has the largest posterior support among these five choices, so it is the grid MAP. Other candidates retain nonzero support.

Watch out

This five-point grid is only for teaching. The real model searches continuous positive abundance values z and uses the Gamma prior, not five equally likely percentages.

12.2 What MAP means

MAP stands for **maximum a posteriori**. It is the parameter value at the highest point of the posterior density:

$$\boxed{\widehat{z}_{\text{MAP}} = \arg\max_{z>0} p(z \mid N, Y, A, \omega, \phi_{\text{ref}}).}$$

Because the logarithm is an increasing function, the same maximum is found by maximizing the log posterior:

$$\widehat{z}_{\text{MAP}} = \arg\max_{z>0} \left[\log p(N \mid z) + \log p(Y \mid N, z) + \log p(z)\right].$$

Big idea

Logs are useful because:

1. multiplication becomes addition, and

2. extremely small probability products are easier for a computer to handle.

12.3 The hill-climbing picture

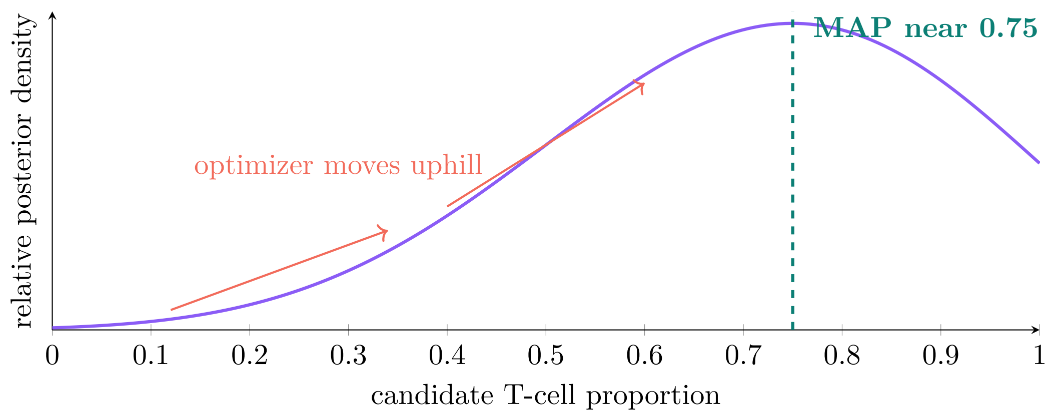


Figure 14: A schematic continuous posterior for the same multinomial example with a flat prior. The optimizer searches for the highest point.

12.4 What MAP does not provide

The primary ShapeMix MVP reports one MAP point estimate. It does *not* report:

- a posterior mean,
- samples from the full posterior,
- a credible interval, or
- calibrated uncertainty for each cell type.

Watch out

Calling a method Bayesian describes how the prior and likelihood define a posterior. Using MAP summarizes that posterior by one point. Do not turn a MAP point into an uncertainty claim it does not support.

13 The full primary model, translated line by line

13.1 First calculate expected contributions

$$\boxed{v_{s,p,b} = \sum_c z_{s,c} A_{c,p} \omega_{c,p,b}}$$

Translate the mathematics

For one spot, peak, and length bin: multiply each cell type's effective amount by its typical total rate and its typical fraction in this bin; then add across cell types.

$$\boxed{\mu_{s,p} = \sum_b v_{s,p,b} = \sum_c z_{s,c} A_{c,p}}$$

Translate the mathematics

Add the expected bin contributions to get the expected total cut-site count. Because each $\omega_{c,p,\cdot}$ sums to one, the length fractions disappear from the total.

$$\boxed{\rho_{s,p,b} = \frac{v_{s,p,b}}{\mu_{s,p}}}$$

Translate the mathematics

Divide one expected bin count by the expected total. The resulting $\rho_{s,p,\cdot}$ values are nonnegative and sum to one.

13.2 Then model the observations

$$\boxed{N_{s,p} \sim \text{NegBinomial}\left(\text{mean} = \mu_{s,p},\ \text{inverse-dispersion} = \max(e_s, \epsilon)\phi_{\text{ref}}\right)}$$

Translate the mathematics

The peak total is a noisy count around $\mu_{s,p}$. Here $e_s = \sum_c z_{s,c}$. The small fixed $\epsilon > 0$ is a numerical safety floor.

$$\boxed{\boldsymbol{Y}_{s,p,\cdot} \mid N_{s,p} \sim \text{Multinomial}\Big(N_{s,p}, \boldsymbol{\rho}_{s,p,\cdot}\Big)}$$

Translate the mathematics

Once the observed total is fixed, split those same cut sites among the three parent-length bins according to the predicted fractions ρ.

13.3 Add the prior and report proportions

$$z_{s,c} \sim \text{Gamma}\left(r_z, \frac{r_z}{m_c}\right), \qquad \pi_{s,c} = \frac{z_{s,c}}{\sum_{c'} z_{s,c'}}.$$

Translate the mathematics

The Gamma prior keeps abundance positive and regularized. Normalize the inferred effective abundances so the reported cell-type proportions sum to one. In the frozen MVP v1, the prior is

Gamma(2, 1) for every cell type.

13.4 Bayes' rule for this exact model

$$\boxed{p(z \mid N, Y, A, \omega, \phi_{\text{ref}}) \propto \underbrace{p(N \mid z, A, \phi_{\text{ref}})}_{\substack{\text{total-count}\\ \text{likelihood}}} \times \underbrace{p(Y \mid N, z, A, \omega)}_{\substack{\text{conditional shape}\\ \text{likelihood}}} \times \underbrace{p(z)}_{\substack{\text{Gamma}\\ \text{prior}}}.}$$

Big idea

The only inferred biological parameter in the primary model is z. A, ω, and ϕ_{ref} are estimated from training reference cells and then held fixed. μ, ρ, e, and π are calculated from those quantities.

14 How to read the advanced ShapeMix tutorial

The destination document is:

`ShapeMix_ATAC_Bayesian_Model_Tutorial.pdf`

Its equations are compact because it assumes most of this companion material. Use three passes instead of trying to master every line at once.

14.1 Pass 1: understand the story

1. Read *What problem is ShapeMix-ATAC solving?*
2. Read *The data table: from peak counts to shape counts.*
3. Read *A short Bayesian warm-up.*
4. Jump to *Worked example: why shape helps.*
5. Finish with *Summary in one page.*

In the current 18-page PDF, these are mainly viewer pages 4–5, 11–12, and 17–18. Section names are more reliable than page numbers if the PDF is rebuilt.

14.2 Pass 2: understand the primary model

Read these sections in order:

1. *Start with the ordinary peak-count model*;
2. *The ShapeMix-ATAC likelihood*;
3. *The generative story*;
4. *Prior and semantics for hidden abundance*;

5. *Complete core model*;
6. *What the primary MAP estimate means*.

Each time an equation appears, mark:

- observed data in gold: N, Y;
- fixed references in blue: $A, \omega, \phi_{\text{ref}}$;
- the unknown in purple: z;
- calculated quantities in teal: e, μ, ρ, π.

14.3 Pass 3: understand scientific safeguards

Then read *Reference signatures*, *Leakage-free experimental structure*, *Diagnostics and ablation tests*, and *Fixed evaluation contract*. These sections explain how the model is tested fairly rather than how Bayes' rule works.

14.4 A five-step equation-reading routine

When an equation looks overwhelming:

1. Identify the left side: what quantity is being defined or modeled?
2. Label every right-side symbol as observed, fixed, inferred, or calculated.
3. Expand one $\sum$ using just two cell types or three bins.
4. Ask whether the line answers "how many?", "how split?", or "what was plausible?"
5. Substitute the small numbers from Section 9.

Check your understanding

You are ready for the advanced tutorial if you can explain why

$$p(N \mid z)\, p(Y \mid N, z)\, p(z)$$

means "total-count fit times conditional-shape fit times prior plausibility."

15 Practice questions

Try these without looking at the answer key. Show the arithmetic, not only the final answer.

1. Seven short, three middle, and two long fragments all have both ends in the same peak. What is the cut-site vector Y, and what is the total N?
2. Can a parent-length bin contain an odd number of cut-site counts in real data? Explain.
3. Two hypotheses have equal priors. Their likelihoods for the observed data are 0.10 and 0.30. What are their normalized posterior probabilities?
4. Explain the difference between $P(\text{short} \mid T)$ and $P(T \mid \text{short})$.
5. If

$$\boldsymbol{Y}_{s,p,\cdot} = (6, 3, 1),$$

what is $N_{s,p}$?

6. A negative-binomial total has $\mu = 20$ and $\phi = 5$. Calculate its variance using the ShapeMix parameterization.
7. Match each job to a distribution:
   (a) noisy total number of cut sites;
   (b) division of a fixed total among three bins;
   (c) prior for a positive effective abundance.
8. In the Section 9 reference,

$$\omega_T = (1/2, 1/3, 1/6), \quad \omega_B = (1/6, 1/3, 1/2).$$

Calculate ρ for 75% T and 25% B.

9. If $z = (3, 1)$, calculate the normalized proportion vector π.
10. Classify each quantity as observed, fixed reference, inferred, or calculated: Y, A, ω, z, μ, and π.
11. What happens to the usefulness of the shape term if every cell type has the same ω?
12. What is the one intentional difference between the count-only and ShapeMix arms?
13. Does a MAP estimate provide a credible interval? What does it provide?
14. Give two reasons for optimizing log probabilities instead of raw probability products.

16 Answer key

1. Each fragment contributes two cut sites:

$$Y = (14, 6, 4), \qquad N = 14 + 6 + 4 = 24.$$

2. Yes. A fragment can contribute only one selected count when one end lands in the peak and the other does not. Both ends inherit the same parent-length bin, but both are not guaranteed to be assigned to the same selected peak.

3. The unnormalized scores are 0.10 and 0.30, totaling 0.40:

$$0.10/0.40 = 0.25, \qquad 0.30/0.40 = 0.75.$$

4. $P(\text{short} \mid T)$ predicts the chance of a short-parent cut when T is known. $P(T \mid \text{short})$ infers the chance of T after short has been observed. Bayes' rule connects them using a prior and normalization.

5.

$$N_{s,p} = \sum_b Y_{s,p,b} = 6 + 3 + 1 = 10.$$

6.

$$\mathrm{Var}(N) = \mu + \frac{\mu^2}{\phi} = 20 + \frac{400}{5} = 100.$$

7. (a) negative binomial; (b) conditional multinomial; (c) Gamma.

8.

$$\begin{aligned}\rho &= 0.75\omega_T + 0.25\omega_B \\ &= \left(\frac{5}{12}, \frac{1}{3}, \frac{1}{4}\right).\end{aligned}$$

At $N = 24$, this corresponds to expected counts $(10, 8, 6)$.

9.

$$\pi = \left(\frac{3}{4}, \frac{1}{4}\right) = (0.75, 0.25).$$

10. Y: observed. A, ω: fixed reference inputs. z: inferred. μ, π: calculated from the inferred and fixed quantities.

11. Shape becomes non-informative about the mixture. Changing z no longer changes the predicted conditional bin fractions ρ.

12. ShapeMix adds only

$$\log p(Y \mid N, z, A, \omega),$$

the conditional multinomial shape term. The total-count likelihood, prior, observations, references, initialization, optimizer, and comparison settings remain matched.

13. No. MAP provides one point: the parameter setting at the highest posterior density found by the optimization.

14. Logs turn products into sums, and they avoid numerical problems caused by multiplying many tiny probabilities.

17 One-page cheat sheet

Four Bayesian words

Prior	Plausibility of unknown values before using the current spot's data.
Likelihood	How well a candidate unknown value predicts the observed data.
Posterior	Updated support after combining likelihood and prior.
MAP	The single unknown value at the highest posterior density.

$$\boxed{\text{posterior} \propto \text{likelihood} \times \text{prior}}$$

Three active distributions

Negative binomial	Noisy peak total N; allows more spread than Poisson.
Conditional multinomial	Allocation of that same fixed N among length bins.
Gamma	Prior on positive effective abundance z.

Core ShapeMix flow

$$\boxed{(z, A, \omega) \longrightarrow (\mu, \rho) \longrightarrow (N, Y)}$$

$$v_{s,p,b} = \sum_c z_{s,c} A_{c,p} \omega_{c,p,b}, \qquad \mu_{s,p} = \sum_b v_{s,p,b}, \qquad \rho_{s,p,b} = \frac{v_{s,p,b}}{\mu_{s,p}}.$$

$$N_{s,p} \sim \mathrm{NegBinomial}(\mu_{s,p}, \max(e_s, \epsilon)\phi_{\mathrm{ref}}), \qquad \boldsymbol{Y}_{s,p,\cdot} \mid N_{s,p} \sim \mathrm{Multinomial}(N_{s,p}, \boldsymbol{\rho}_{s,p,\cdot}).$$

$$z_{s,c} \sim \mathrm{Gamma}\left(r_z, \frac{r_z}{m_c}\right), \qquad \pi_{s,c} = \frac{z_{s,c}}{\sum_{c'} z_{s,c'}}.$$

For frozen MVP v1 evaluation, $r_z = m_c = 2$, so this prior is $\mathrm{Gamma}(2, 1)$ for every cell type.

Roles

Observed	Y, N
Fixed from reference	$A, \omega, \phi_{\mathrm{ref}}$
Inferred by MAP	z
Calculated	e, μ, ρ, π

Big idea

Count-only asks how many cut sites occurred. ShapeMix asks the same question and also asks how those same cuts were divided among parent-fragment length bins.

Final readiness checklist

You are ready to read the advanced PDF if you can:

- distinguish $P(E \mid H)$ from $P(H \mid E)$;
- explain prior, likelihood, posterior, and MAP;
- identify $Y, N, A, \omega, z, \mu, \rho, \phi, \pi$;
- explain the jobs of the negative binomial, multinomial, and Gamma;
- show why $p(N)\, p(Y \mid N)$ is not double-counting; and
- explain why the MVP's MAP output is a point estimate, not a credible interval.

Relationship to the project documents

This companion is an educational guide to the advanced tutorial at:

```
docs/ShapeMix/tutorials/
ShapeMix_ATAC_Bayesian_Model_Tutorial.pdf
```

The executable model semantics are governed by `docs/ShapeMix/model_specification.md`. The examples here deliberately use small numbers and simplified one-peak settings; they teach the model's logic but are not study results.